

Task-Arithmetic Model Merging for the *clearsky* Project

A Historical Log of the Merging Experiments, Their Rationale and Findings

Experimental report

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1 Context and Setup

The *clearsky* project restores degraded astronomical images with a conditional diffusion model. A single **ImprovedUNet** (a U-Net with self-attention, sinusoidal time embeddings, GroupNorm/SiLU blocks) is trained as a **v-prediction DDPM** with a cosine noise schedule. Three separate single-task *experts* were fine-tuned from one shared pre-trained base θ_0 :

- **star_removal** — removal of stars from the field;
- **image_restoration** — repair of dead / hot pixels;
- **super_resolution** — upsampling / detail recovery.

The goal of every experiment below is the same: obtain *one* network that handles all three degradations at once, without retraining, using **task arithmetic**. For each expert one computes a *task vector* $\tau_t = \theta_t - \theta_0$, and the merged weights are

$$\theta_{\text{merged}} = \theta_0 + \sum_t \lambda_t \tau_t,$$

where the per-task coefficients λ_t are chosen by a search over a validation set. The five notebooks are a chronological record of how the *search strategy* and, more importantly, the *selection metric* were refined as flaws in earlier choices became visible. The network architecture, the v-prediction objective and the merging formula itself never change; everything that follows is about *how the λ 's are chosen and judged*.

A naming caveat carried through the log. The dataset folders and the weight names do *not* share the same abbreviations, which caused a real bug that experiment 4 had to fix. The convention finally adopted is: folder **SR** $\rightarrow$ star_removal, folder **SU** $\rightarrow$ super_resolution, folder **IR** $\rightarrow$ image_restoration. Combined-degradation folders (**SR_IR**, **SR_SU**, **IR_SU**, **SR_IR_SU**) hold images corrupted by several degradations simultaneously.

Throughout, the task vector norms were stable across runs ($\|\tau_{\text{SR}}\| \approx 56.4$, $\|\tau_{\text{IR}}\| \approx 50.3$, $\|\tau_{\text{SU}}\| \approx 80.9$), i.e. the super-resolution expert moved much further from the base than the other two — an early hint that it would dominate any naive combination.

2 Summary of the Five Experiments

Table ?? condenses the whole progression. The central story is the migration of the *selection metric*: from raw MSE (Exp. 1–2), to MSE normalised against per-task baselines (Exp. 3–4), to a perceptual, standardised Z-score (Exp. 5). Each change was forced by a specific failure of the previous metric.

Table 1: Historical evolution of the merging experiments. Winner reported as λ_{SR} , λ_{IR} , λ_{SU} = coefficients. “score” meaning differs per row (see the metric column).

Exp.	Search strategy	Selection metric	λ grid	Winner (SR/IR/SU)	Headline result
1	Single point (confirmation run)	Raw MSE, pooled per-task val splits	1 point	0.9/0.1/0.0	val_loss = 0.01237; SU dropped
2	Two-stage grid (safe zone $\rightarrow$ fine balance)	Raw MSE per folder, averaged	27 combos	1.0/0.1/0.0	val_loss = 0.00398; SU still 0
3	Single-stage grid	Normalised MSE/expert; PSNR, SSIM	≤ 4 combos	(run truncated)	Introduces fair, scale-free metric
4	Two-stage grid; folder $\rightarrow$ task bug fixed	Norm. vs <i>expert</i> (stage 1) & vs θ_0 (stage 2)	36 combos	0.9/0.1/0.1	score = 0.1091; SU now non-zero
5	Two-stage grid + diagnostics	Z-score: LPIPS on SU, MSE on SR/IR	8 combos	0.9/0.1/0.1	Reveals perceptual interference on SU

3 Experiment 1 — Baseline Merge and Raw-MSE Evaluation

What it does

The first notebook establishes the full pipeline: load θ_0 and the three expert checkpoints (from the `ema_model` key), check shape/key compatibility, build the task vectors, and evaluate a merged model on isolated per-task validation splits selected through a `split_manifest` JSON. Evaluation samples are produced with a fast **DDIM** sampler (25 steps) and scored by **mean squared error** against ground truth; the per-task MSEs are pooled into a single `val_loss`. A final, full-quality **DDPM** pass re-checks the chosen model.

Rationale

This is the minimum viable merging loop — it answers “does the arithmetic even produce a usable network?” As shipped, the λ grid holds a *single* point, $\lambda_{\text{SR}}=0.9$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.0$: the notebook is a confirmation run of a coefficient set found in an earlier exploratory sweep (a code comment records a prior best of $\lambda_{\text{SR}}=0.5$, $\lambda_{\text{IR}}=0.6$, $\lambda_{\text{SU}}=0.0$ at `val_loss` 0.00553).

Findings

The merged model reaches `val_loss` = 0.012368, with a very uneven per-task breakdown: `star_removal` MSE $\approx 7 \times 10^{-5}$ but `image_restoration` ≈ 0.0268 and `super_resolution` ≈ 0.0222 . **Super-resolution is switched off entirely** ($\lambda_{\text{SU}} = 0$). This is the first symptom of the problem that dominates the rest of the log: because SU carries by far the largest absolute MSE, any pooled-MSE objective is minimised by *suppressing* SU rather than by balancing the three tasks.

4 Experiment 2 — Two-Stage Grid Search

What changed

- A genuine **grid search** replaces the single point: $\lambda_{\text{SR}} \in \{0.8, 0.9, 1.0\}$, $\lambda_{\text{IR}} \in \{0.1, 0.3, 0.5\}$, $\lambda_{\text{SU}} \in \{0.0, 0.1, 0.3\}$ — 27 combinations.

- The search is split into **two stages**. *Stage 1* (“safe zone”) scores every combination on the single-degradation folders IR/SR/SU and keeps the best 10. *Stage 2* re-scores only those 10 on the combined-degradation folders (SR_IR, SR_SU, IR_SU, SR_IR_SU) to pick the final balance.
- The dataset class is rewritten to pool several roots and take a **deterministic random subsample** (5% per folder), because the full sets are too large; the deterministic seed keeps every DDP process aligned on the same slice.

Rationale

Exp. 1 only ever tested hand-picked points. The two-stage design encodes a reasonable intuition: first find coefficients that each task tolerates in isolation (a “safe zone”), then break ties using the harder, realistic combined-degradation images — while keeping the combinatorics affordable by subsampling.

Findings

Stage 1’s best is $\lambda_{\text{SR}}=0.9$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.0$ (val_loss 0.002975); stage 2’s best is $\lambda_{\text{SR}}=1.0$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.0$ (val_loss 0.003978). The machinery is richer, but the **metric is still raw averaged MSE**, so the outcome is the same pathology as Exp. 1: $\lambda_{\text{SU}} = 0$ **wins again**. The averaged loss is dominated by the SU folder (its MSE ~ 0.006 dwarfs IR/SR at ~ 0.001), so the optimiser is rewarded for ignoring super-resolution. This makes explicit that the *search* was never the bottleneck — the *objective* was.

5 Experiment 3 — A Fair, Normalised Metric

What changed

- **Normalised selection metric.** Instead of raw MSE, the score is

$$\text{score} = \frac{1}{|T|} \sum_t \frac{\text{MSE}_{\text{merge}}[t]}{\text{MSE}_{\text{expert}}[t]},$$

i.e. each task’s error is measured *relative to its own expert*. This is invariant to the very different MSE scales/floors of the three tasks and no longer collapses onto whichever task has the largest absolute error.

- **Expert baselines** are computed once, up front (each expert evaluated only on its own task).
- **One loader per task** (no pooling), and a **fixed per-task seed**: the same initial noise is used for the expert and for every λ combination, so comparisons are fair.
- **Perceptual metrics** PSNR and a dependency-free **SSIM** are added to the report; results are dumped to JSON.

Rationale

This is the conceptual turning point. Exp. 1–2 proved that averaging raw MSE systematically discards the highest-error task. Normalising each task by its own expert puts the three tasks on a common footing — a merge that reaches, say, $1.2\times$ its expert’s error on every task is now clearly preferable to one that is perfect on two tasks and absent on the third. PSNR/SSIM were added because MSE alone says little about perceived image quality.

Findings

The logged run is truncated (only the `image_restoration` expert baseline prints before the notebook moves on), so this experiment contributes the *method* rather than a final winner. Its ideas — normalisation, one-off expert baselines, fixed seeds, perceptual reporting — are inherited by every later experiment.

6 Experiment 4 — Two-Stage Search with Normalised Metrics

What changed

- **Unifies** Exp. 2’s two-stage search with Exp. 3’s normalised metric, and **fixes the folder→task mapping bug**: the two earlier scripts had contradicted each other on whether SR meant `star_removal` or `super_resolution`. The correct pairing (`SR→star_removal`, `SU→super_resolution`, `IR→image_restoration`) is fixed here, since *every* stage-1 baseline depends on pairing each expert with the right folder.
- **Two baselines**. Stage 1 normalises each single-degradation folder by its *expert*. Stage 2’s combined folders have no single expert, so they are normalised by θ_0 (the un-merged base model) instead.
- The grid grows and shifts: $\lambda_{\text{SR}} \in \{0.7, 0.8, 0.9, 1.0\}$, $\lambda_{\text{IR}} \in \{0.1, 0.3, 0.5\}$, $\lambda_{\text{SU}} \in \{0.1, 0.3, 0.5\}$ (36 combos). Crucially $\lambda_{\text{SU}} = 0$ is **removed** from the grid, forcing a genuine three-task compromise.

Rationale

Combine the best of both prior threads. Normalising the combined folders by θ_0 answers “how much does merging improve on doing nothing?”—the natural reference when no single expert exists for a mixed degradation.

Findings

The fair metric changes the answer for the first time. Stage 1’s winner is $\lambda_{\text{SR}}=0.9$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.1$ (score 0.7512) — **super-resolution is finally non-zero**. Stage 2 confirms the same coefficients (score 0.1091; the merged model beats θ_0 on the combined folders by large margins, e.g. ratios 0.03–0.31). Expert baselines were $\text{MSE}_{\text{SR}} = 0.00094$, $\text{MSE}_{\text{IR}} = 0.00244$, $\text{MSE}_{\text{SU}} = 0.00842$, confirming SU’s much higher error floor. Notably the winning merge’s SR ratio is ~ 1.17 (slightly worse than the star-removal expert) while IR and SU ratios are *below* 1 — the normalised metric is now trading tasks off against each other, exactly as intended.

7 Experiment 5 — Perceptual Z-Score and Interference Diagnosis

What changed

- **LPIPS** (AlexNet backbone, via the `lpips` library) is added as a perceptual metric and computed on any folder involving `super_resolution`; **SSIM is dropped**.
- The MSE/baseline ratio is replaced by a **combined Z-score**: for each folder, the relevant metric is standardised ($z = (x - \mu)/\sigma$) across all candidate merges and averaged. Folders containing SU are ranked by **LPIPS**; SR/IR folders by **MSE**. Lower is better.
- A sampling-free **deterministic validation loss** is added (the v-prediction loss at fixed timesteps with fixed noise), giving a stable proxy that does not depend on the stochastic sampler.

- Efficiency: DDIM cut to **10 steps**, evaluation capped at 16 samples/folder; the full ancestral `sample()` method is removed. Stage-1 and final models are saved separately.
- A dedicated **diagnostics** cell measures task-vector norms and pairwise cosine similarities and compares θ_0 , the pure SU expert, the winning merge and an “SU-pushed” variant at 10 vs. 50 DDIM steps.

Rationale

Exp. 4 gave a balanced-looking answer, but visual inspection of super-resolution outputs did not match the low MSE. **MSE rewards blur**: a smooth, detail-free prediction can score well on MSE while looking clearly worse. LPIPS was introduced precisely to catch this, and the Z-score makes the different metrics comparable across folders so LPIPS (for SU) and MSE (for SR/IR) can be combined into one ranking.

Findings

The Z-score again selects $\lambda_{\text{SR}}=0.9$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.1$, but the diagnostics deliver the log’s most important result. On the SU folder:

Model (on SU)	Det. val loss	MSE	LPIPS
θ_0 (pure pre-train)	0.0835	0.00827	0.2303
SU fine-tuned ($\lambda_{\text{SU}}=1$)	0.0880	0.00893	0.2203
Winning merge (0.9/0.1/0.1)	0.1557	0.00707	0.4080

The winning merge has the *lowest* MSE (0.00707) yet by far the *worst* LPIPS (0.408 vs. ~ 0.22 for both the pure base and the pure SU expert). Two conclusions follow:

1. **MSE was actively misleading.** The whole MSE-based line of search was rewarding a perceptually *worse*, blurrier merged model — vindicating the move to LPIPS.
2. **This is task-arithmetic interference, not a sampling artifact.** The LPIPS gap persists at 50 DDIM steps (0.409), so it is not blur from cheap sampling. Since the SU expert *alone* is fine (LPIPS 0.22) but the merge is not, the super-resolution capability is being destroyed by interference from the other task vectors when they are summed.

The cosine similarities support this reading: the task vectors are nearly orthogonal ($\cos \approx 0.08\text{--}0.12$), which is normally favourable for task arithmetic, but $\|\tau_{\text{SU}}\| \approx 80.8$ is much larger than the others, so even a small mixing coefficient on SR/IR perturbs the SU direction enough to wreck perceptual quality. The notebook’s own reading points the next steps at interference-aware merging — **TIES** / **DARE**, renormalising τ_{SU} , or using an earlier (fewer-epoch) SU checkpoint.

8 Overall Trajectory

The five experiments are best read not as five different models but as a single, sharpening question: *what does it mean for a merged diffusion model to be “good” at three tasks at once?*

1. **Exp. 1–2** answered it with raw MSE and were repeatedly told that the best three-task model is a *two*-task model ($\lambda_{\text{SU}} = 0$) — a metric artifact, not a real result.
2. **Exp. 3–4** fixed the metric by normalising against per-task baselines, and immediately recovered a genuine three-task compromise ($\lambda_{\text{SR}}=0.9$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.1$).
3. **Exp. 5** showed that even a good MSE-based answer hides a perceptual failure on super-resolution, replaced MSE with LPIPS where it matters, and diagnosed the cause as task-vector interference — turning a hyperparameter search into a concrete research direction.

The concrete deliverable stabilised on the coefficient set $\lambda_{\text{SR}}=0.9$, $\lambda_{\text{IR}}=0.1$, $\lambda_{\text{SU}}=0.1$, but the durable finding is methodological: for merging generative restoration models, **the choice of evaluation metric matters more than the choice of search algorithm**, and pixel-wise error must be supplemented by a perceptual metric before any merge can be trusted.